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```
clc;
clear;
close all;

% FIR Filter Coefficients
b = [1 2 3 2 1];
a = 1;

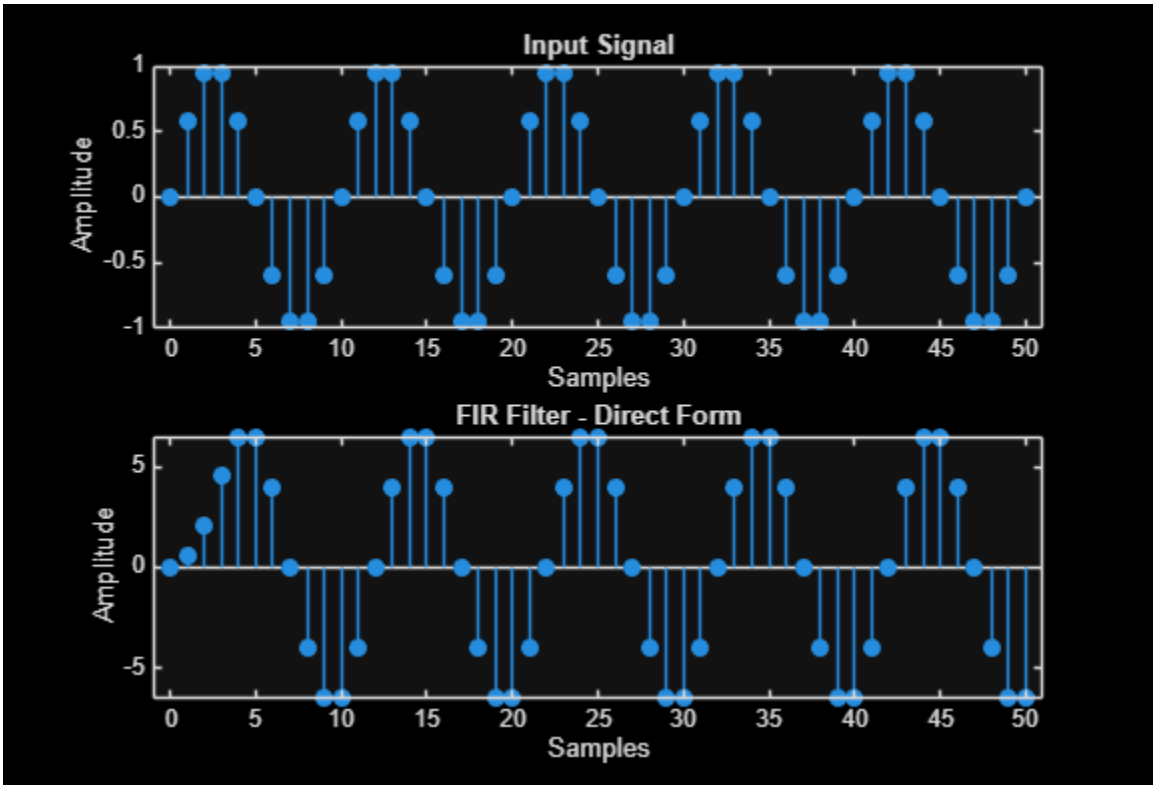
% Input Signal
n = 0:50;
x = sin(0.2*pi*n);
```

1. Direct Form Realization

```
y_direct = filter(b, a, x);

figure;
subplot(2,1,1);
stem(n, x, 'filled');
title('Input Signal');
xlabel('Samples');
ylabel('Amplitude');

subplot(2,1,2);
stem(n, y_direct, 'filled');
title('FIR Filter - Direct Form');
xlabel('Samples');
ylabel('Amplitude');
```



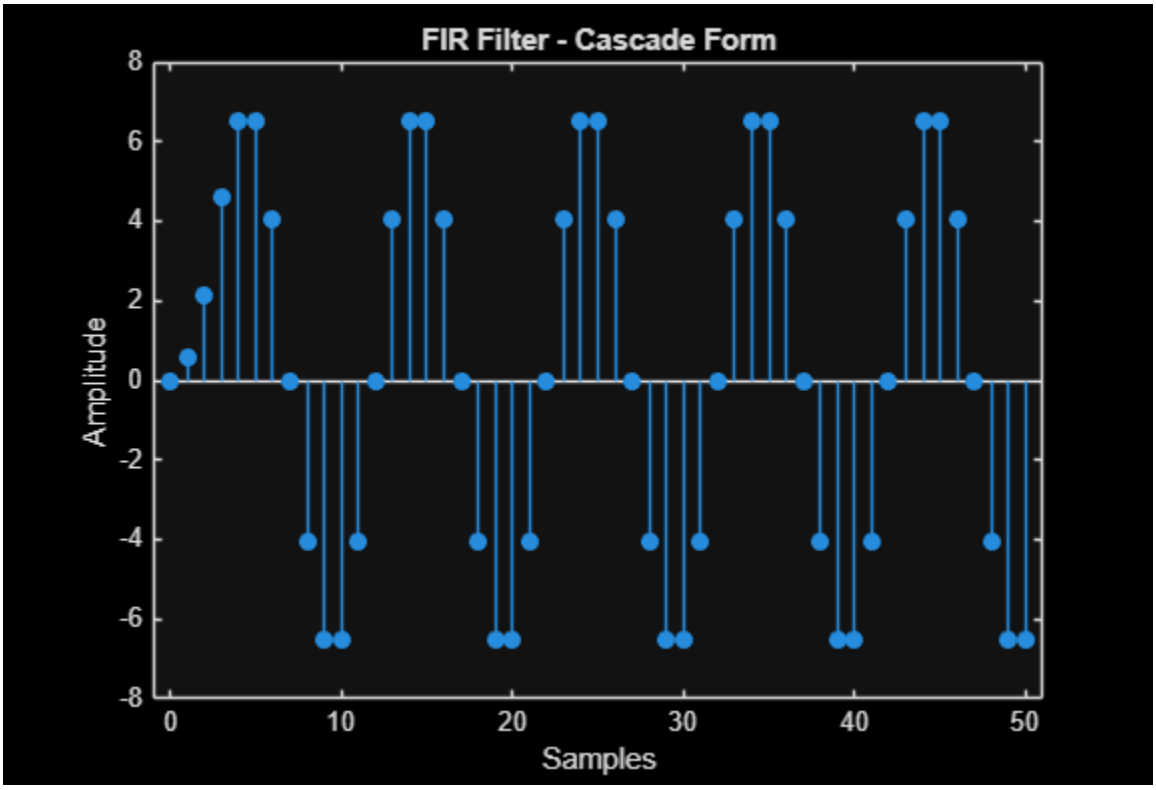
2. Cascade Form Realization

Factor the FIR transfer function into sections

```
[sos, g] = tf2sos(b, a);

y_cascade = g * x;
for k = 1:size(sos,1)
    y_cascade = filter(sos(k,1:3), ...
        sos(k,4:6), y_cascade);
end

figure;
stem(n, y_cascade, 'filled');
title('FIR Filter - Cascade Form');
xlabel('Samples');
ylabel('Amplitude');
```



3. Linear-Phase FIR Structure

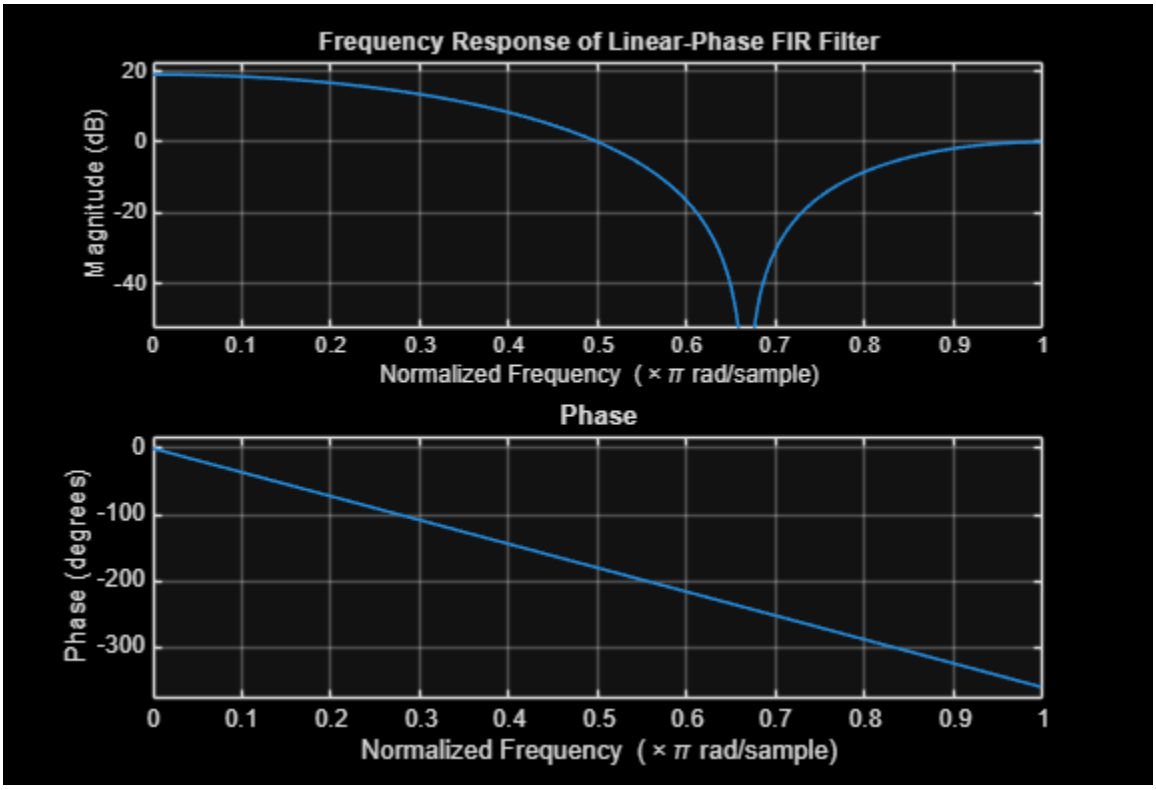
Symmetric coefficients give linear phase

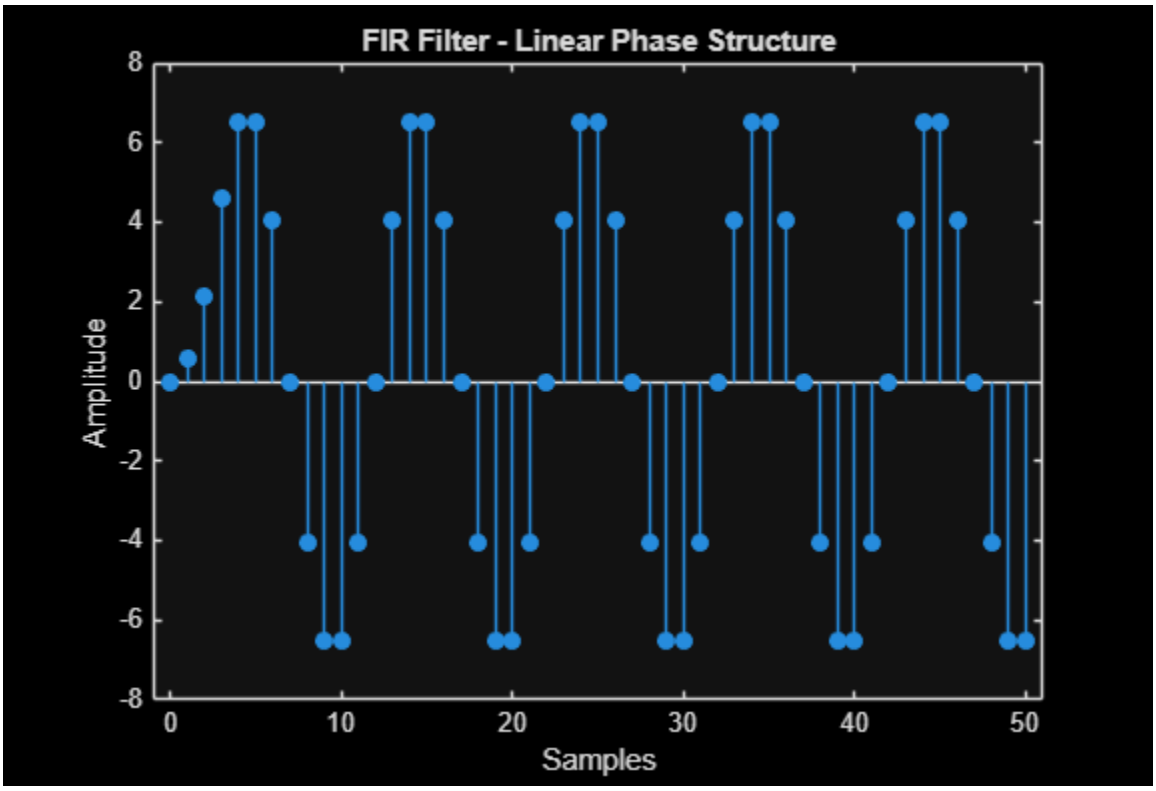
```
b_lp = [1 2 3 2 1];

y_linear = filter(b_lp, 1, x);

figure;
freqz(b_lp, 1, 512);
title('Frequency Response of Linear-Phase FIR Filter');

figure;
stem(n, y_linear, 'filled');
title('FIR Filter - Linear Phase Structure');
xlabel('Samples');
ylabel('Amplitude');
```


Frequency Response of Linear-Phase FIR Filter
Magnitude (dB)
20
0
-20
-40
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-100
-200
-300
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)


FIR Filter - Linear Phase Structure
Amplitude
8
6
4
2
0
-2
-4
-6
-8
0
10
20
30
40
50
Samples

Display Results

```
disp('FIR Filter Coefficients:');
disp(b);

disp('Direct Form Output:');
disp(y_direct);

disp('Cascade Form Output:');
disp(y_cascade);

disp('Linear Phase Output:');
disp(y_linear);
```

```
FIR Filter Coefficients:
     1     2     3     2     1

Direct Form Output:
  Columns 1 through 7

         0    0.5878    2.1266    4.6165    6.5186    6.5186    4.0287

  Columns 8 through 14

    0.0000   -4.0287   -6.5186   -6.5186   -4.0287   -0.0000    4.0287

  Columns 15 through 21

    6.5186    6.5186    4.0287    0.0000   -4.0287   -6.5186   -6.5186

  Columns 22 through 28

   -4.0287   -0.0000    4.0287    6.5186    6.5186    4.0287    0.0000

  Columns 29 through 35

   -4.0287   -6.5186   -6.5186   -4.0287   -0.0000    4.0287    6.5186

  Columns 36 through 42

    6.5186    4.0287    0.0000   -4.0287   -6.5186   -6.5186   -4.0287

  Columns 43 through 49

   -0.0000    4.0287    6.5186    6.5186    4.0287    0.0000   -4.0287

  Columns 50 through 51

   -6.5186   -6.5186

Cascade Form Output:
  Columns 1 through 7
```

```
         0    0.5878    2.1266    4.6165    6.5186    6.5186    4.0287

  Columns 8 through 14

   -0.0000   -4.0287   -6.5186   -6.5186   -4.0287    0.0000    4.0287

  Columns 15 through 21

    6.5186    6.5186    4.0287   -0.0000   -4.0287   -6.5186   -6.5186

  Columns 22 through 28

   -4.0287    0.0000    4.0287    6.5186    6.5186    4.0287   -0.0000

  Columns 29 through 35

   -4.0287   -6.5186   -6.5186   -4.0287    0.0000    4.0287    6.5186

  Columns 36 through 42

    6.5186    4.0287   -0.0000   -4.0287   -6.5186   -6.5186   -4.0287

  Columns 43 through 49

   -0.0000    4.0287    6.5186    6.5186    4.0287    0.0000   -4.0287

  Columns 50 through 51

   -6.5186   -6.5186

Linear Phase Output:
  Columns 1 through 7

         0    0.5878    2.1266    4.6165    6.5186    6.5186    4.0287

  Columns 8 through 14

    0.0000   -4.0287   -6.5186   -6.5186   -4.0287   -0.0000    4.0287

  Columns 15 through 21

    6.5186    6.5186    4.0287    0.0000   -4.0287   -6.5186   -6.5186

  Columns 22 through 28

   -4.0287   -0.0000    4.0287    6.5186    6.5186    4.0287    0.0000

  Columns 29 through 35

   -4.0287   -6.5186   -6.5186   -4.0287   -0.0000    4.0287    6.5186

  Columns 36 through 42

    6.5186    4.0287    0.0000   -4.0287   -6.5186   -6.5186   -4.0287
```

```
  Columns 43 through 49

   -0.0000    4.0287    6.5186    6.5186    4.0287    0.0000   -4.0287

  Columns 50 through 51

   -6.5186   -6.5186
```

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